Tyler LaBonte

Email: tlabonte@gatech.edu
Website: https://tyler-labonte.com

Employment

MICROSOFT RESEARCH AI FRONTIERS — Remote
Senior Researcher — 2026–

MICROSOFT RESEARCH AI FRONTIERS — Redmond, WA
Machine Learning Research Intern — 2025

- Joined core development team of a 15 billion parameter multimodal vision-language model.
- Led integration effort, tool-use training, and evaluation for a multimodal computer-use agent application.
- Investigated grounding capabilities of our model on fine-grained visual reasoning tasks.

Education

GEORGIA INSTITUTE OF TECHNOLOGY — 2021–2026
Ph.D., Machine Learning

UNIVERSITY OF SOUTHERN CALIFORNIA — 2017–2021
B.S., Applied and Computational Mathematics, *magna cum laude*

Skills: Python, PyTorch, TensorFlow, HuggingFace, DeepSpeed, W&B, Linux CLI, Kubernetes, Docker, Git, Vim, LaTeX

Selected Publications

1. Phi-4-reasoning-vision-15B Technical Report
   ($\alpha\beta$) Jyoti Aneja, Michael Harrison, Tyler LaBonte, John Langford, and Eduardo Salinas
   Technical Report, 2026.
2. Task Shift: From Classification to Regression in Overparameterized Linear Models
   Tyler LaBonte*, Kuo-Wei Lai*, and Vidya Muthukumar (* denotes equal contribution)
   AISTATS 2025
3. The Group Robustness is in the Details: Revisiting Finetuning under Spurious Correlations
   Tyler LaBonte, John C. Hill, Xinchen Zhang, Vidya Muthukumar, and Abhishek Kumar
   NeurIPS 2024
4. Towards Last-layer Retraining for Group Robustness with Fewer Annotations
   Tyler LaBonte, Vidya Muthukumar, and Abhishek Kumar
   NeurIPS 2023
5. Scaling Novel Object Detection with Weakly Supervised Detection Transformers
   Tyler LaBonte, Yale Song, Xin Wang, Vibhav Vineet, and Neel Joshi
   WACV 2023

Selected Service & Awards

Reviewing: CVPR, TMLR, ICLR, ICML, NeurIPS, CVPR Workshop on Demographic Diversity in Computer Vision, ICLR Workshop on Spurious Correlations and Shortcut Learning

DoD National Defense Science and Engineering Graduate Fellowship ($170,000) — 2021

NSF Graduate Research Fellowship ($138,000—declined) — 2021